RegenRL: Optimizing Electric Vehicle Braking Torque Split via Soft Actor-Critic

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GitHub: `https://github.com/zerokelvinrahul/RegenRL`

Abstract

Electric vehicles recover braking energy through their traction motor, but regeneration is constrained by motor back-EMF cutoffs, battery BMS charge acceptance, and road-slope axle stability, while the driver demands consistent deceleration. We formulate the continuous regenerative-to-friction braking split as a reinforcement learning problem and solve it with Soft Actor-Critic (SAC). Over 20 unseen test drive cycles, 3-seed SAC outperforms a validation-tuned fixed-ratio controller and an automotive-style rule-based heuristic. Against the rule-based controller it gains +22.1% reward (paired $p = 0.0026$) with 5% more recovered energy and a 52% lower mean stopping-distance violation ($p = 0.13$, a trend rather than a significant effect). SAC closes 47.3% of the reward gap between the rule-based controller and a greedy hardware-limit reference.

1. Problem and RL Formulation

At every 0.1 s step the agent chooses the fraction α_t of the driver's torque demand T_t^{dem} to be delivered by regeneration; the rest, $(1-\alpha_t)T_t^{dem}$, goes to friction brakes. Requesting too little regen wastes energy, while requesting more than the hardware can deliver (low speed, high SoC, high power) leaves a torque deficit and a longer stopping distance.

State space $\mathcal{S} \subset \mathbb{R}^4$ (normalized):

$$s_t = \left[\tfrac{v_t}{35},\ \mathrm{SoC}_t,\ \tfrac{\theta_t}{10^\circ},\ \tfrac{T_t^{dem}}{2500}\right],$$

with speed $v \in [0, 35]$ m/s, battery state of charge, road slope $\theta \in [-10^\circ, 10^\circ]$ and driver torque demand in Nm.

Action space $\mathcal{A} = [0, 1]$ (continuous): the regen fraction α_t, giving the requested regen torque $T_t^{req} = \alpha_t T_t^{dem}$.

Reward. A dense three-term reward balances efficiency, safety and comfort:

$$r_t = w_e E_t - w_v\left(d_t^{viol} + 0.05\tfrac{T_t^{def}}{1000}\right) - w_s(\alpha_t - \alpha_{t-1})^2, \quad (1)$$

where E_t is the net energy stored in the battery (kJ) after motor efficiency and I^2R losses, $d_t^{viol} = v_t\,(a_t^{short}/g)\,\Delta t$ is the stopping-distance violation caused by the deceleration shortfall a_t^{short}, T_t^{def} is the torque deficit (Nm), and $(w_e, w_v, w_s) = (1.5, 14, 0.15)$. The large w_v makes safety dominant, so energy is harvested only when it does not compromise braking performance.

2. Environment and Algorithm

2.1. Simulation Environment

We built a custom Gymnasium environment (`EVBrakingEnv`) for a 1650 kg compact EV. Each episode consists of four braking segments (urban and highway mix) with randomized initial speed, target speed, slope, peak torque demand (600–2400 Nm) and initial SoC (0.50–0.95). The achieved regen torque is $T^{act} = \min(T^{req}, T^{cap})$, where $T^{cap} = \min(T^{motor}, T^{batt})$ combines: (i) a low-speed back-EMF fade (zero below 1.2 m/s, full above 4 m/s), (ii) the motor constant-torque / constant-power curve (1850 Nm, 60 kW), (iii) a slope-dependent axle stability factor, and (iv) a BMS charge-power taper above 82% SoC. Deceleration follows from friction plus regen torque and aerodynamic, rolling and grade forces. Stored energy accounts for a motor efficiency map (0.72–0.92) and pack I^2R losses, and updates SoC.

2.2. Algorithm: Soft Actor-Critic

Because α_t is continuous, we use SAC [1, 2], an off-policy maximum-entropy actor-critic method that maximizes $\mathbb{E}\big[\sum_t r_t + \beta\mathcal{H}(\pi(\cdot|s_t))\big]$. The actor outputs a Gaussian squashed by tanh and rescaled to $[0, 1]$. Two critics (clipped double-Q) with soft target updates ($\tau = 0.005$) reduce overestimation, and the temperature β is tuned automatically (target entropy -1). Networks have two 256-unit ReLU layers, trained with Adam (learning rate 3×10^{-4}), $\gamma = 0.99$, batch size 256, a 100k replay buffer, 1k random warm-up steps and 15k environment steps per seed (seeds 42, 43, 44). The environment follows the Gymnasium API [3].

2.3. Baselines

Fixed-Ratio: static $\alpha = 0.5$, the best constant from a validation sweep over $[0.1, 1.0]$ (seeds 500–519). **Rule-Based:** $\alpha = 0.9$, linearly tapered below 8 m/s and above 80% SoC, mimicking automotive production logic. **Greedy reference (oracle):** $\alpha_t = \min(1.0, T_t^{cap}/T_t^{dem})$ using exact instantaneous hardware limits. It yields zero torque deficit and serves as an empirical upper bound, though it ignores smoothness and efficiency-map effects. All controllers are evaluated on identical test seeds.

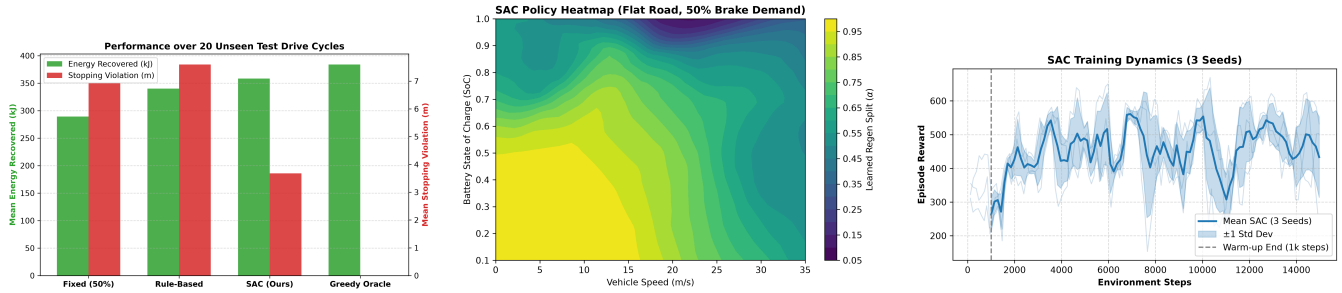


Figure 1: Left: mean energy recovered and stopping violation over 20 test cycles (SAC averaged over 3 seeds). Middle: learned policy $\alpha(v, \text{SoC})$ of the seed-42 SAC agent on flat road (50% brake demand). Right: SAC learning curves over 3 seeds with $\pm 1\sigma$ band.

Table 1: Test performance (mean ± std over 20 test cycles; best practical method in bold; greedy reference is an empirical upper bound).

Method	Reward	Energy (kJ)	Viol. (m)
Fixed (50%)	325.5 ± 267.2	289.2 ± 114.0	6.92 ± 7.45
Rule-Based	392.4 ± 215.4	340.2 ± 211.6	7.60 ± 8.73
SAC (3 seeds)	**479.2** ± 292.5	**358.2** ± 167.4	**3.68** ± 3.02
Greedy ref.	575.8 ± 269.7	384.0 ± 179.8	0.00 ± 0.00

3. Results

Table 1 reports test metrics over 20 unseen cycles (seeds 200–219); SAC metrics are averaged over the 3 trained seeds per cycle. Because cycle difficulty varies with speed and slope, we use paired Student's t-tests on per-cycle differences Δ = SAC − Baseline. Relative to the rule-based heuristic, SAC improves mean reward by +22.1% (+86.78, 95% CI [34.5, 139.1], paired $t = 3.472$, $p = 0.0026$, win rate 80%) while recovering 5% more energy (358.2 vs. 340.2 kJ). Mean stopping violation falls 52% (from 7.60 m to 3.68 m), but this difference alone is not significant ($p = 0.13$; SAC is lower in 11 of 20 cycles). Relative to fixed 50%, SAC yields +47.2% reward (+153.75, $t = 10.429$, $p < 10^{-4}$, win rate 100%). Against the greedy reference (575.8 reward, 0 m violation), SAC closes 47.3% of the gap between the rule-based controller and the reference, while having to learn the hardware-limit map from reward alone. Across the three training seeds, mean test reward is 479.2 ± 5.4 and energy 358.2 ± 3.0 kJ.

4. Discussion

Learned behaviour. The trajectory (Fig. 2) shows that SAC modulates α dynamically instead of holding a constant split: its α varies between roughly 0.4 and 0.9 and qualitatively follows the greedy reference, dipping where the reference dips, whereas the fixed controller stays at 0.5 and the rule-based heuristic stays near 0.9 throughout. Requesting 0.9 even when the hardware cap is lower leads to torque shortfalls and stopping violations. The policy heatmap (Fig. 1) shows α varying smoothly with speed and SoC, with the SoC dependence reflecting the BMS taper above 82%.

Training. Reward improves quickly after the 1k-step warm-up and then fluctuates, since episodes differ in difficulty (Fig. 1); the three seeds reach similar test performance (479.2 ± 5.4 reward).

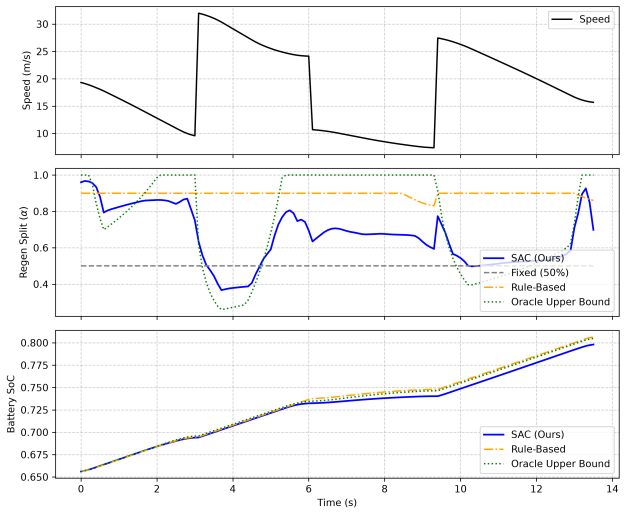


Figure 2: Speed, regen split α and SoC over one cycle (seed 101, a held-out seed outside the 200–219 test set; seed-42 SAC agent).

Limitations. Variance across cycles reflects profile difficulty, which the paired tests isolate. The violation reduction versus rule-based is not statistically significant at the 5% level. Current simplifications include independent braking maneuvers (speed resets at segment boundaries), a single-axle lumped driveline, and a 1 kWh effective buffer window. Future work includes front/rear brake-by-wire distribution, cell thermal modeling, and continuous traffic driving schedules.

References

[1] T. Haarnoja, A. Zhou, P. Abbeel, S. Levine. Soft Actor-Critic: Off-Policy Maximum Entropy Deep Reinforcement Learning with a Stochastic Actor. *ICML*, 2018.

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[3] M. Towers et al. Gymnasium: A Standard Interface for Reinforcement Learning Environments. *arXiv:2407.17032*, 2024.